

Roll No.

TPE-303

B. TECH. (PE) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
FLUID AND PARTICLE MECHANICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Explain compressibility and bulk modulus. (CO1)

OR

- (b) Explain how does a fluid reacts to shear stress. How is it different from a solid ? (CO1)

2. (a) What are body and surface forces ? Is gravity a body force and surface force ? (CO1)

OR

- (b) What is Pascal's law ? Derive with appropriate diagram. (CO1)

3. (a) What is hydrostatic law ? Using this law explain how will the pressure change with depth in a pond. (CO1 and CO2)

P. T. O.

(2)

OR

- (b) With the help of a diagram explain how the gauge pressure be measured using the U-type manometer ? (CO1 & CO2)
4. (a) What are the Lagrangian and Eulerian methods for studying the fluid flow ? (CO2)

OR

- (b) What is unsteady flow ? How is it different from non-uniform flow ? (CO2)
5. (a) Explain continuity equation for the case of flow in a pipe with variable cross-section. (CO2)

OR

- (b) Under what conditions are the Euler equations of motion valid ? (CO2)